



**TECHNICAL REPORT ON STUDENT INDUSTRIAL WORK
EXPERIENCE SCHEME**

[SIWES]

BY

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AT

PETIRABLE ELECTRIC POLES LTD

KM 2 GBERIGBE – IMOTA ROAD, IKORODU, LAGOS STATE.

**IN PARTIAL FULFILMENT FOR THE AWARD OF BACHELOR OF
ENGINEERING [B.ENG]**

EKITI STATE UNIVERSITY, ADO EKITI, EKITI STATE.

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CERTIFICATION PAGE

I hereby certify that this report of SIWES was prepared and compiled by **Ogungbesan Victor Ayodeji** with matriculation number **210504002** from the **Department of Mechanical Engineering, Faculty of Engineering, Ekiti State University, Ado-Ekiti, Ekiti State**. For the completion of his **three months** SIWES undertaken at **PETIRABLE ELECTRIC POLES LTD, KM 2 GBERIGBE – IMOTA ROAD, IKORODU, Lagos**.

STUDENT TRAINEE: OGUNGBESAN Victor Ayodeji (210504002)

SIGNATURE AND DATE:

UNIVERSITY-BASED SUPERVISOR: ENGR. OBASANYA

SIGNATURE AND DATE:

EKSU SIWES COORDINATOR:

SIGNATURE AND DATE:

DECLARATION

I hereby declare that this comprehensive report was compiled by me (Ogungbesan Victor Ayodeji) and entails precisely what I have done during my SIWES undertaken at PETIRABLE ELECTRIC POLES LTD at KM 2 GBERIGBE -IMOTA ROAD, IKORODU, LAGOS STATE. I declare that this report and its content have not been submitted to this or any other institution of learning for the purpose of consummating the requisites for the award of any degree.

DEDICATION

I dedicate this report to my mom, siblings and in honour of my late dad.

ACKNOWLEDGEMENT

I express my sincere gratitude to God Almighty for his guidance and protection throughout the period of my SIWES training.

I also appreciate both my university-based supervisor (Engr. Obasanya) and Industrial-based supervisor (Engr Olabode Irabor) for their support and contributions to the success of this training.

I acknowledge with thanks and deep appreciation for the support and values I got from my mother, sisters, friends, family and relatives, staff of Petirable Electric Poles and to everyone who in one way or the other contributed to the success of my SIWES training but whose names I have not included. I say thank you and God bless.

ABSTRACT

This report presents is a summary of the experience I acquired during my three months students' industrial work experience scheme at PETIRABLE ELECTRIC POLES LTD, KM 2 GBERIGBE-IMOTA ROAD, IKORODU. With highlights majorly on mechanical engineering, giving full enlightenment on the equipment utilised on tasks such installation etc while giving a clear idea of the student involvement in all operational activities carried out in the production firm.

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CHAPTER ONE

1.0 BACKGROUND OF STUDY [MECHANICAL ENGINEERING]

Mechanical engineering is one of the oldest and the broadest branches of engineering, rooted in the principles of physics, mathematics, material science. It emerged from the need to design, analyse and improve machines and mechanical systems that converts energy into useful work. From the era of the industrial revolution to the modern age of automation and advanced manufacturing, mechanical engineering has remained central to technological development. The discipline focuses on the application of principles such as thermodynamics, mechanics, heat transfer and fluid dynamics. With rapid technological development the field has evolved to incorporate modern tools such as CAD, CAM.

The discipline plays a crucial role in key sectors including construction, power generation.

1.1 INTRODUCTION TO SIWES

The SIWES (Student Industrial Working Experience Scheme) is an integral part of the course structure in all Polytechnics and Universities in Nigeria, to help students acquire necessary practical experience to complement their course of study in the institution and prepare them professionally to perform effectively in the chosen field of study.

1.2 HISTORY OF SIWES

The Student Industrial Working Experience Scheme (SIWES) started as result of the established ITF in 1973. This came from the government degree No. 47 of 8th October 1971 constitution as amended in 1990, which highlighted the capacity of building human resources in industry. Commerce and government through training and retaining of workers to effectively provide the much-needed high quality goods and services in a dynamic economy like ours. The growing concern of industrialists, the graduate of institution in higher level lack adequate practical background studies preparatory for employment in industries led to the formation of SIWES by ITF in 1993 / 1999.

SIWES is a skill-training program designed to expose and prepare the students of Agriculture, Engineering, Science and Technology, Medical Science and Pure and Applied Science for the Industrial work situation, which they are likely to meet after graduation. The duration of SIWES is 4 months in polytechnics at the end of ND 1, 4 months in College of Education at the end of NCE II and 6 months in universities at the end of 300, 400, 500 levels depending on the discipline.

1.3 VISION AND MISSION STATEMENT OF SIWES

VISION

To produce skilled, competent graduates through effective industrial training and practical exposure.

MISSION

To bridge the gap between theory and practice by exposing students to real work environments and relevant industrial skills

1.4 AIMS AND OBJECTIVES OF SIWES

- i. To provide avenues for students to acquire Industrial skills and experience during their course of study.
- ii. To prepare students for the Industrial work situation they are likely to meet after graduation.
- iii. To expose students to work methods and techniques in handling equipment and machinery.
- iv. To provide students with the opportunities to apply their educational knowledge in real work situations, hereby bridging the gap between theory and practice.
- v. To make the transition of schooling to world of work easier through enhancing the students' contact for later job placement.

1.5 IMPORTANCE OF SIWES TO MY COURSE OF STUDY

It provides hands on experience that complements classroom theory, improves technical skills, exposes me to real industry practices and prepares me for professional work after graduation.

CHAPTER TWO

2.0 BRIEF HISTORY OF THE ESTABLISHMENT

PETIRABLE ELECTRIC POLES LTD, formerly known as Rehoboth, was founded in 2013 by Engineer Olabode Irabor. The company's head office is situated at KM 2 Gberigbe - Imota, Ikorodu, Lagos State.

Petirable Electric Poles Ltd. is a registered and certified electric pole manufacturer and power services provider in Nigeria company that has operated across all 36 states in Nigeria, with a remarkable track record in the production of standard concrete electric poles and power service operations across the country.

With a vision rooted in innovation, integrity, and reliability, Petirable stands as a trusted name in Nigeria's growing power and infrastructure industry.

2.1 ORGANIZATION CORE VALUES

- **Uncompromising Quality:** Every product and service is delivered to meet highest standard.
- **Customer Satisfaction:** We listen, advice and provide tailored solutions for our clients' needs.
- **Prompt Delivery:** We are known for our exceptional delivery speed once payment is confirmed.

2.2 PETIRABLE MISSION AND VISION

VISION

To become the number one pole manufacturing and power solutions company in Nigeria and across Africa.

MISSION

To bring light to every community by providing reliable, high-quality electric poles and power solutions.

2.3 PETIRABLE SERVICES

Petirable offers 3 distinct services that has served over 200 clients which includes individuals and communities, government agencies and multinational corporations such as : Ikeja Electric, MTN, IBEDC. These services includes:

a. Petirable Electric Poles Services

Petirable is dedicated to the production of world-class concrete electric poles. The organisation specialises in producing durable, high-quality, and standard concrete electric poles designed to meet national and international standards. Their poles, which are SON (Standard Organisation of Nigeria) certified are engineered to withstand Nigeria's and Africa's diverse environmental conditions while ensuring safety and reliability in power distribution.

Their range of products include:

- 6.5-meter Standard Poles
- 8-meter Standard Poles
- 28ft Low Tension (LT) Poles

Special Poles:

- 6.5-meter Standard Poles
- 8-meter Standard Poles
- 28ft Low Tension (LT) Poles

Every Petirable pole goes through strict quality control and is SON-approved, guaranteeing strength, stability, and long-lasting performance.

b. Petirable Power Services

This arm of the organisation focuses on power infrastructure installation, logistics, and consultancy. They provide comprehensive services that support construction, installation, and maintenance for both public and private clients.

Our services include:

- Transformer Installation & Repair
- Electrical Installation Consulting
- Technical & Cable Assistance
- Excavation & Digging: Professional groundwork for pole planting and power installations.
- Expert Advisory & Consultancy Service

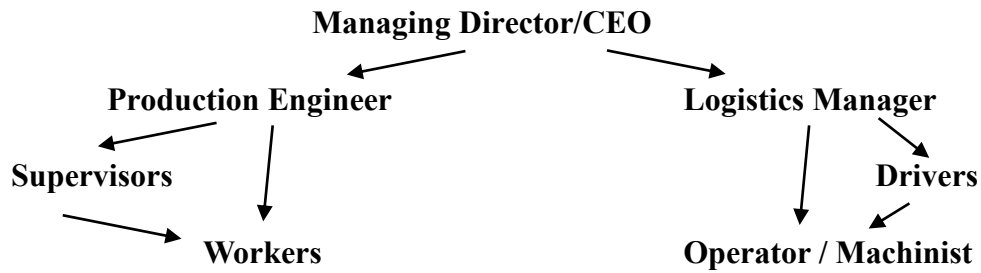
Petirable Power Services Limited ensures seamless project execution, focusing on efficiency, safety, and client satisfaction.

c. Petirable Hiab Services

As a proud member of the Hiab & Crane owners of Nigeria, the organisation offers the following Hiab services:

- Lifting and transportation of generators and heavy equipment
- Container loading, unloading, and dismantling
- Pole installation lifting assistance
- Machinery movement and repositioning
- Site clearing and logistics support
- Project-based equipment mobilisation
- Any other lifting and transport service is achievable with a Hiab crane.

2.4 ORGANIZATION STRUCTURE



2.5 JUSTIFICATION FOR THE CHOICE OF INDUSTRY

The company was chosen because it provided practical exposure to industrial machines such as the Hiab use for haulage services. These machines offered hands on experience in areas related to machine operation and workshop safety, which are core aspects of the mechanical engineering training.

In addition, I chose the industry to learn more about the niche of production engineering, an area in which I have developed strong interest and passion, knowing fully well that the company would provide adequate exposure.

CHAPTER THREE

3.0 DESCRIPTION OF WORK DONE (POLE PRODUCTION)

3.1 COMMON TERMS USED IN POLE PRODUCTION

- **Batching:** This is the process of measuring and combining raw materials - cement, sand, aggregate, water, and sometimes admixtures - in the correct proportions to produce concrete of required strength. Incorrect batching leads to weak or brittle poles
- **Mixing:** The mechanical or manual blending of batched materials until a uniform concrete mixture is achieved.
- **Moulding:** This is the process of shaping fresh concrete by placing it into a mould that defines the pole's size, length, taper, and cross-section.

A mould is a specially designed steel form used to shape the concrete pole. It is reusable.

- **Reinforcement (Rebar / Steel Cage):** Steel bars or wires placed inside the mould before casting to improve the pole's tensile strength and resistance to bending.
- **Casting:** This is the act of pouring mixed concrete into the mould containing reinforcement.
- **Compaction:** This is the removal of air pockets from fresh concrete using vibration or tamping. This is done to prevent honeycombing and to increase strength.
- **Curing:** The process of maintaining moisture and temperature to allow concrete to gain strength over time.

This can be done using various methods such as water sprinkling, wet covering (jute bags) or steam curing. The curing process usually takes 7 – 28 days depending on pole specification.

- **Demoulding:** This is the removal of the hardened pole from the mould after sufficient initial strength is achieved.

3.2 POLE PRODUCTION PROCESS

During my training, I was exposed to the processes involved in the production of concrete poles. The processes involved are listed below:

1. **Design and Speciation:** Before production begins, the pole design is determined based on specifications such as pole length, reinforcement details (number and diameter of steel rods) and concrete grade.
2. **Preparation of Moulds:** Next, steel moulds are cleaned thoroughly to remove old concrete residues. Mould surfaces are coated with mould oil to prevent concrete from sticking. The mould is assembled and tightened properly to avoid leakage during casting. Openings for fittings (like bolt holes or cross-arm holes) are fixed in position.
3. **Cutting and preparation of reinforcement:** Steel reinforcement bars are cut to meet the required length. Main longitudinal bars are arranged evenly. Stirrups or binding wires alongside the use of spacing technique are used to hold the bars in position and ensure structural integrity.
4. **Placement of Reinforcement Cage in the Mould:** The reinforcement cage is placed centrally inside the mould. Alignment is checked to ensure uniform strength and balance.
5. **Concrete Batching and Mixing:**

Materials used:

- Cement
- Fine aggregate (sand)
- Coarse aggregate (granite)
- Clean water

Materials are batched accurately according to the mix design. Mixing is done using a concrete mixer until a uniform mix is achieved.

6. **Casting of Concrete:** Fresh concrete is poured into the mould gradually. Filling is done in layers to avoid segregation. The concrete completely covers the reinforcement cage. Care is taken to prevent air pockets and voids.

7. **Compaction of Concrete:** Proper compaction is done to remove trapped air and improve surface finish.
8. **Mould removal:** After sufficient hardening, the mould is carefully opened, and the pole is removed without causing damage.
9. **Curing:** Curing is done and lasts for days depending on specifications
10. **Finishing and Marking:** Sharp edges are smoothed. Surface defects are repaired. Identification marks are written or stamped. Identification marks include Manufacturer's name, Date of production, Pole length and type
11. **Quality control and testing:** Visual inspection for cracks or deformities. A rebound hammer is used to assess the quality of the hardened poles
12. **Storage:** Produced poles are stored well properly.

3.3 POLE ERECTION PROCESS

1. **Site survey:** In this process, locations for pole installations are assessed, distances are measured, obstacles are identified and other factors that influence installation process are considered.
2. **Pole Pit Excavation:** Holes are dug to receive the pole. This depends on the type of poles to be installed. For example, for low tension poles, holes of 4 – 5 ft are dug.
3. **Pole Transportation:** Poles are moved from the production site to installation site using the Hiab / Truck.
4. **Pole Erection:** The pole is lifted and positioned vertically inside the hole, backfilled with soil or concrete and then compacted to stabilize the pole.

3.4 SAFETY RULES AND REGULATIONS

- a. Personal protective equipment such as gloves, reflective jacket, helmet, safety boots, safety goggles were worn at all times during work activities
- b. Hands and clothes were kept away from the moving parts of machineries.
- c. All equipment and machinery were operated under supervision.
- d. Ensured that all tools and equipment were kept in good working conditions and maintenance.



Fig. 1 and 2: Images depicting the process involved in the use of Hiab in pole erection.

3.5 CHALLENGES ENCOUNTERED AND PROPOSED SOLUTIONS

During SIWES activities, several challenges were encountered which posed risks to safety, productivity, and overall work efficiency. These challenges include electrocution, accidents, and harassment from thugs. Each challenge and its corresponding solutions are discussed below.

1. **Electrocution:** This was a major risk during the installation of poles carried out under live power lines. Lack of adequate safety awareness increased the risk of electric shock which could have resulted in severe injury or death.

Proffer solutions

- a. Switch off power sources before the planting of poles especially under live power lines.
 - b. Use of personal protective apparatus such as gloves, Safety boots, Reflective jacket, Helmet etc.
2. **Accident:** Accidents could occur due to factors such as improper handling of tools and machinery, poor maintenance of crane lifting equipment which can lead to injuries etc

Proffered solution

- a. Provide and enforce the use of PPE
- b. Ensure workers are properly trained on the safe use of tools and equipment
- c. Carry out regular safety inspections and risks assessments
- d. Supervise work activities closely to ensure compliance with safety procedures

3. Harassment from thugs: Harassment from thugs was experienced during site operations. This posed serious threat to the safety of life and property causing fear, disruption of work and ultimately assault.

Proffered solutions

- a. Employ security personnel to safeguard workers and equipment
- b. Establish emergency communication channels for quick response during threats
- c. Inform local authorities and community leaders before commencing work in high-risk areas
- d. Schedule work activities for the planting of poles during safer hours of the day.

CHAPTER FOUR

4.0 EXPERIENCE GAINED AND RELEVANCE TO COURSE OF STUDY

I gained hands on experience on the use of the Hiab; it is highly relevant to my course of study as it is an example of a real-life application of ergonomics as I was taught in production engineering. I could apply the concepts of certain topics such as materials science and engineering, maintenance engineering, measurement and instrumentation, and workshop practices to solve real life problems.

I was exposed to real life engineering practices, workplace discipline and the application of engineering principles to solve practical problems.

I gained technical skills on the use of heavy-duty trucks for various usage, ranging for carriage and haulage services. I was exposed to the various manufacturing stages involved in the production of poles and the importance of the materials selected for its making.

Overall, I was exposed to the real-life application of engineering principles and theories.

4.1 RECOMMENDATIONS

1. **For Students:** Students should approach SIWES with seriousness and eagerness to learn.
2. **For Institutions:** Institutions should strengthen supervision
3. **For Industries:** Industries should provide adequate training facilities
4. **For Government and ITF:** Improved funding and timely payment of allowances.

4.2 CONCLUSION

During the training period, Relevant skills were acquired in equipment handling, Safety practices, teamwork and problem solving, all of which are essential to the mechanical engineering profession.

Overall, the SIWES programme was a worthwhile experience that contributed significantly to my academic and professional development.